Chemiluminescence of Luminol

Purpose

To observe the light produced by the oxidation reaction of luminol and study and explain its chemical properties and reactions.

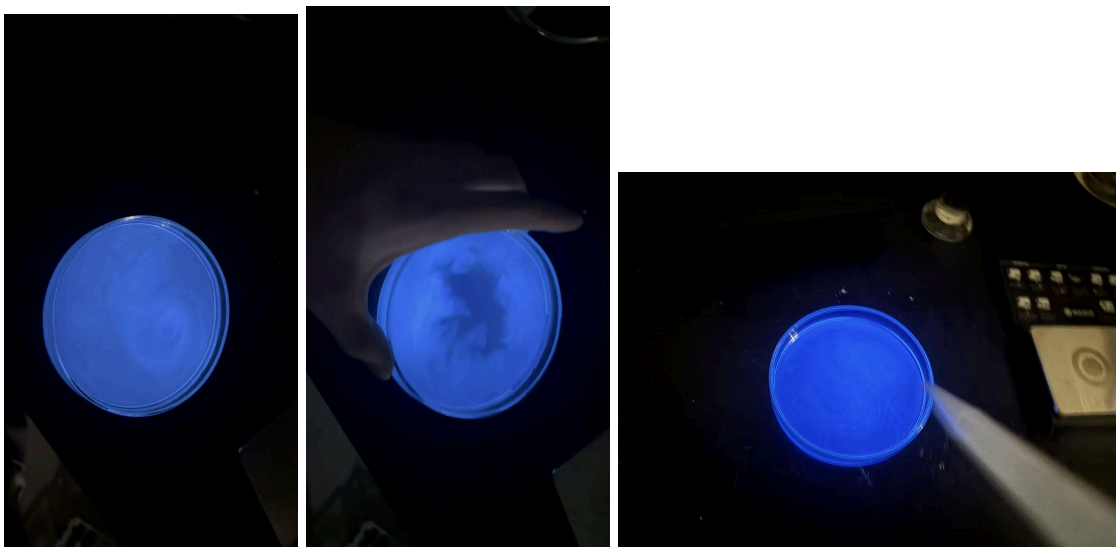
Materials

Chemical	Quantity
Luminol	0.010 g
Anhydrous sodium carbonate, Na_2CO_3	0.20 g
Sodium bicarbonate, $NaHCO_3$	1.20 g
Commercial ammonium carbonate / Baker's ammonia	0.025 g
Copper(II) sulfate pentahydrate, $CuSO_4 \cdot 5H_2O$	0.020 g
3% hydrogen peroxide, H_2O_2	2.5 mL
Distilled water	To prepare 50 mL of each solution

Procedure

- 30ml of distilled water was used to dissolve the sodium carbonate.
- Luminol added with sodium bicarbonate, ammonium carbonate and copper sulfate.
- 2.5 ml of 3% hydrogen peroxide was diluted with water to value of 50ml
- Mixing them together.

Observation

- A bright blue light was visible in the container
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Chemistry explanation

- The main reason for generating light during he reaction is that the excited product releases energy in the form of light once it returns to the ground state.
- The carbonate solution provides the alkaline condition in which luminol loses protons and forms a dianion. In the reaction, hydrogen peroxide acts as the oxidizing agent, and copper helps the reaction proceed faster.
- The oxidation involves peroxide intermediates that experience bond rearrangements, releasing nitrogen gas and producing 3- aminophthalate, and the energy comes from electronic excitation.
- This glow will weaken once the energy is all released; it is irreversible because the structure of the luminol has changed, with no reverse electron transfer.